finessed. In his novel, he describes the wormhole through which Eleanor Arroway traveled as now being maintained by exotic matter, but as having been created in the distant past by some infinitely advanced civilization, from which all records have been lost.

We physicists, of course, are not happy to relegate wormhole creation to prehistory. We want to know whether and how the Universe's topology can be changed *now*, within the confines of physical law.

We can imagine two strategies for constructing a wormhole where before there was none: a *quantum strategy*, and a *classical strategy.*

The quantum strategy relies on *gravitational vacuum fluctuations* (Box 12.4), that is, the gravitational analogue of the electromagnetic vacuum fluctuations discussed above: random, probabilistic fluctuations in the curvature of space caused by a tug-of-war in which adjacent regions of space are continually stealing energy from each other and then giving it back. Gravitational vacuum fluctuations are thought to be everywhere, but under ordinary circumstances they are so tiny that no experimenter has ever detected them.

Just as an electron's random degeneracy motions become more vigorous when one confines the electron to a smaller and smaller region (Chapter 4), so also gravitational vacuum fluctuations are more vigorous in small regions than in large, that is, for small wavelengths rather than for large. In 1955, John Wheeler, by combining the laws of quantum mechanics and the laws of general relativity in a tentative and crude way, deduced that in a region the size of the *Planck–Wheeler length*,[6] 1.62×10^{-33} centimeter or smaller, the vacuum fluctuations are so huge that space as we know it "boils" and becomes a froth of quantum foam—the same sort of quantum foam as makes up the core of a spacetime singularity (Chapter 13; Figure 14.3).

Quantum foam, therefore, is everywhere: inside black holes, in interstellar space, in the room where you sit, in your brain. But to see the quantum foam, one would have to zoom in with a (hypothetical) supermicroscope, looking at space and its contents on smaller and smaller scales. One would have to zoom in from the scale of you and me (hundreds of centimeters) to the scale of an atom (10^{-8} centimeter), to the scale of an atomic nucleus (10^{-13} centimeter), and then on downward by *twenty* factors of 10 more, to 10^{-33} centimeter. At all the early, "large" scales,